

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

AI-Driven Crop Disease Prediction and Precision Farming Assistant

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Problem Statement

Agriculture is rapidly evolving by adopting Artificial Intelligence (AI), drones, Internet of Things (IoT) sensors, and computer vision technologies to enhance crop productivity, optimize resource management, and promote long-term agricultural sustainability. However, a major bottleneck in traditional farming is that farmers typically identify crop diseases only after advanced, visible symptoms appear on plants, which leads to irreversible crop damage, severely reduced yields, and an over-reliance on chemical pesticides. Traditional monitoring techniques depend heavily on manual field inspections and consultation with specialized agricultural experts, who are often scarce or inaccessible in remote rural regions, resulting in delayed diagnoses and unoptimized farming practices. To overcome these challenges, an AI-driven precision farming assistant is needed to continuously analyze leaf and crop images alongside real-time hyper-local weather conditions and IoT soil parameters to detect crop diseases at an early, manageable stage. By automatically diagnosing early-stage pathologies, recommending targeted treatment measures, calculating optimized precision irrigation schedules, and providing field-level crop health analytics, the system eliminates unnecessary chemical usage, conserves critical agricultural resources, and fundamentally supports sustainable farming practices.

1. Study and Scenario Analysis

1.1 Business Domain & Industry Context

The proposed system operates within the **AgTech (Agricultural Technology)** and Precision Agriculture domain. Modern agriculture is rapidly transitioning toward data-driven cultivation models by integrating Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT) sensors, and hyper-local weather analytics. The core objective of this ecosystem is to enhance crop productivity, optimize resource management, mitigate environmental impacts, and foster long-term agricultural sustainability.

1.2 Primary & Secondary Stakeholders

- Farmers / Field Operators (Primary Users): Capture crop images, receive diagnostic results, access precision treatment plans, and monitor irrigation schedules.
- Agronomists / Agricultural Extension Officers: Validate flagged AI diagnostic reports, update regional disease advisories, and provide expert consultation for complex cases.
- System Administrators: Oversee system operations, manage user roles, monitor AI model accuracy/drift, and maintain API integrations.
- Agri-Input Partners / Suppliers: Receive diagnostic telemetry (where authorized) to supply certified organic treatments, biological controls, and precision fertilizers.

1.3 Existing Challenges in Agriculture

- Delayed Disease Identification: Traditional inspection relies on manual visual checks. Diseases are frequently detected only after advanced visible symptoms emerge, leading to severe yield loss.
- Overuse of Chemical Pesticides: In the absence of precise diagnostic tools, farmers often resort to broad-spectrum chemical sprays, inflating costs and degrading soil and environmental health.
- Scarcity of Expert Advisory: Specialized agronomists are often unavailable in remote rural regions, delaying critical treatment decisions.

- **Unoptimized Irrigation & Inputs:** Irrigation and fertilization schedules are typically executed using guesswork rather than real-time soil parameters and meteorological predictions.

1.4 System Objectives

- **Early Disease Detection:** Leverage deep learning computer vision to diagnose crop diseases at early asymptomatic or initial-symptom stages.
- **Sustainable Treatment Recommendations:** Provide actionable, eco-friendly treatment measures emphasizing biological controls and targeted input usage.
- **Precision Irrigation Scheduling:** Dynamic scheduling calculated from real-time IoT soil parameters and weather forecast data.
- **Comprehensive Crop Health Analytics:** Deliver field-level analytical dashboards tracking crop health metrics and historical risk trends.

2. Identification of Requirements

2.1 Functional Requirements (FR)

Req ID	Feature / Function	Requirement Description
FR-01	User Management & Authentication	Secure user registration, profile management, and field location/crop type mapping.
FR-02	Crop Image Upload & Capture	Mobile and web capture interface for high-resolution imagery of leaves, stems, or fruits.
FR-03	AI Crop Disease Prediction	Automated computer vision inference engine providing disease identification and confidence scores.
FR-04	IoT & Weather Data Ingestion	Real-time stream ingestion of soil parameters (moisture, pH, NPK) and local weather forecasts.
FR-05	Precision Treatment Advisor	Generation of targeted, sustainable, and eco-friendly treatment plans and chemical/organic dosages.

FR-06	Smart Irrigation Scheduler	Automated calculation of irrigation volume and timing based on soil moisture and precipitation forecasts.
FR-07	Crop Health Analytics	Interactive dashboard displaying spatial health maps, historical disease trends, and yield predictions.
FR-08	Agronomist Escalation	Workflow to escalate low-confidence AI predictions or severe outbreaks to human agronomists.

2.2 Non-Functional Requirements (NFR)

- **Performance:** AI disease diagnosis must complete within < 3 seconds over standard 3G/4G cellular networks.
- **Usability:** Intuitive, multi-lingual, and icon-driven UI specifically engineered for users with varying digital literacy levels.
- **Reliability & Availability:** 99.5% operational uptime during core cultivation seasons, supported by offline image capture capability.
- **Scalability:** Cloud-native architecture capable of processing thousands of concurrent image analysis requests during peak morning hours.
- **Security & Data Privacy:** Role-Based Access Control (RBAC), end-to-end encryption (TLS 1.3) for telemetry, and farmer location privacy protection.

3. Actors and Use Case Modeling

3.1 Identification of System Actors

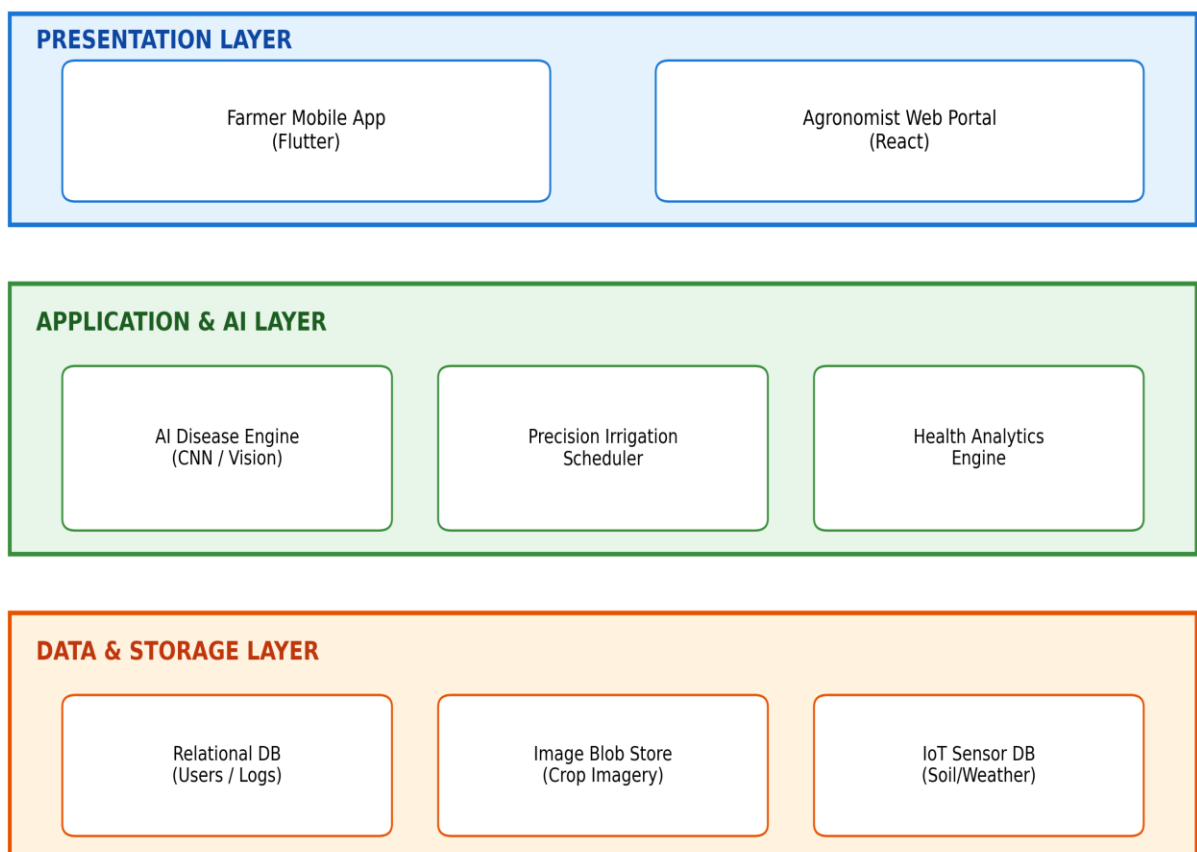
- **Farmer (Primary Human Actor):** Interacts with the system to capture crop images, view diagnostics, follow irrigation schedules, and request expert support.
- **Agronomist (Secondary Human Actor):** Reviews escalated cases, validates AI output, and updates treatment knowledge bases.
- **IoT Sensors / Weather Service (System Actor):** Automated external systems feeding real-time soil metrics and atmospheric data.

- System Administrator (System Actor): Manages application configurations, user accounts, and AI model deployments.

3.2 Major Use Cases

- UC-01: Upload Crop Image for Diagnosis
- UC-02: View Disease Prediction & Sustainable Advisory
- UC-03: Access Precision Irrigation Schedule
- UC-04: Escalate Case to Agronomist
- UC-05: Ingest Soil Telemetry & Weather Forecast
- UC-06: System Maintenance & Model Retraining

3.3 Structural Use Case Representation



4. Software Requirement Specification (SRS) Details

4.1 System Scope

The AI-Driven Precision Farming Assistant encompasses an end-to-end digital agriculture framework. It connects field-level mobile data capture with cloud AI inference microservices, IoT sensor stream processing, dynamic weather API integration, and expert agronomy validation workflows. The software promotes eco-friendly practices by prioritizing organic remedies and precise input dosages.

4.2 Detailed Use Case Description: UC-01 Early Disease Prediction

Use Case Name:	UC-01 Upload Crop Image & Early Disease Prediction
Primary Actor:	Farmer
Pre-conditions:	User is authenticated; mobile camera permissions granted; field location and crop type registered.
Main Success Scenario:	1. Farmer captures or uploads photo of leaf/crop. 2. Mobile client compresses image and transmits to Cloud Inference Service. 3. Convolutional Neural Network (CNN) runs inference for pathology. 4. System cross-references local soil moisture and weather parameters. 5. System returns disease diagnosis, severity percentage, confidence score, and eco-friendly treatment advice.
Alternative Flow:	If AI confidence score is below 70%, the system automatically prompts the farmer to escalate the case to an Agronomist (UC-04).
Post-conditions:	Diagnostic log recorded in farm analytics history; local alerts generated if contagious.

4.3 External Interface Requirements

- User Interface: Cross-platform mobile/web interface with high-contrast UI elements, voice commands, icon-guided workflows, and support for regional languages.
- Hardware Interface: Wireless communication protocols (MQTT / HTTP REST) connecting IoT soil sensor nodes and gateways to the cloud ingestion engine.
- Software Interface: Integrations with global weather APIs (OpenWeatherMap/NOAA) and cloud Machine Learning deployment services (TensorFlow Serving / ONNX Runtime).

5. Assumptions and Constraints

5.1 Assumptions

- Farmers possess basic camera-enabled smartphones capable of running standard web or hybrid mobile applications.
- Deployed IoT field sensors are properly calibrated and operational within the monitored farm plots.
- Sufficiently annotated regional crop disease image datasets are available for machine learning model training.

5.2 System Constraints

- Technical Constraint: Intermittent cellular connectivity in rural regions necessitates offline image caching and queued synchronization.
- Operational Constraint: Treatment recommendations must strictly comply with local agricultural regulations and sustainable organic standards.
- Economic Constraint: Infrastructure deployment and cloud API operational costs must remain low to ensure affordable access for smallholder farmers.
- Time Constraint: Development, testing, and SRS delivery must adhere strictly to academic/project scheduling windows.

6. Requirement Validation and Consistency

6.1 Completeness & Traceability Verification

All stated business challenges from the problem scenario have been systematically mapped to functional features:

- Challenge: Late Disease Detection -> Addressed by FR-02 (Image Upload) and FR-03 (AI Vision Diagnosis).
- Challenge: Excessive Pesticide Usage -> Addressed by FR-05 (Sustainable Precision Treatment Advisor).
- Challenge: Manual Inspection Dependency -> Addressed by FR-03 (Automated AI) and FR-08 (Agronomist Escalation Workflow).
- Challenge: Unoptimized Resource Usage -> Addressed by FR-04 (IoT Stream) and FR-06 (Smart Irrigation Scheduler).

6.2 Quality Improvement Recommendations

- Voice-Guided Interactive Assistant: Integrate speech recognition and audio response for non-literate farmers.
- Geospatial Community Outbreak Alerts: Automatically issue spatial warning notifications to nearby registered farms when a contagious disease is verified in the region.
- Closed-Loop Feedback Reinforcement: Collect farmer treatment feedback (e.g., crop recovery status) to continuously retrain and refine AI diagnostic models.